

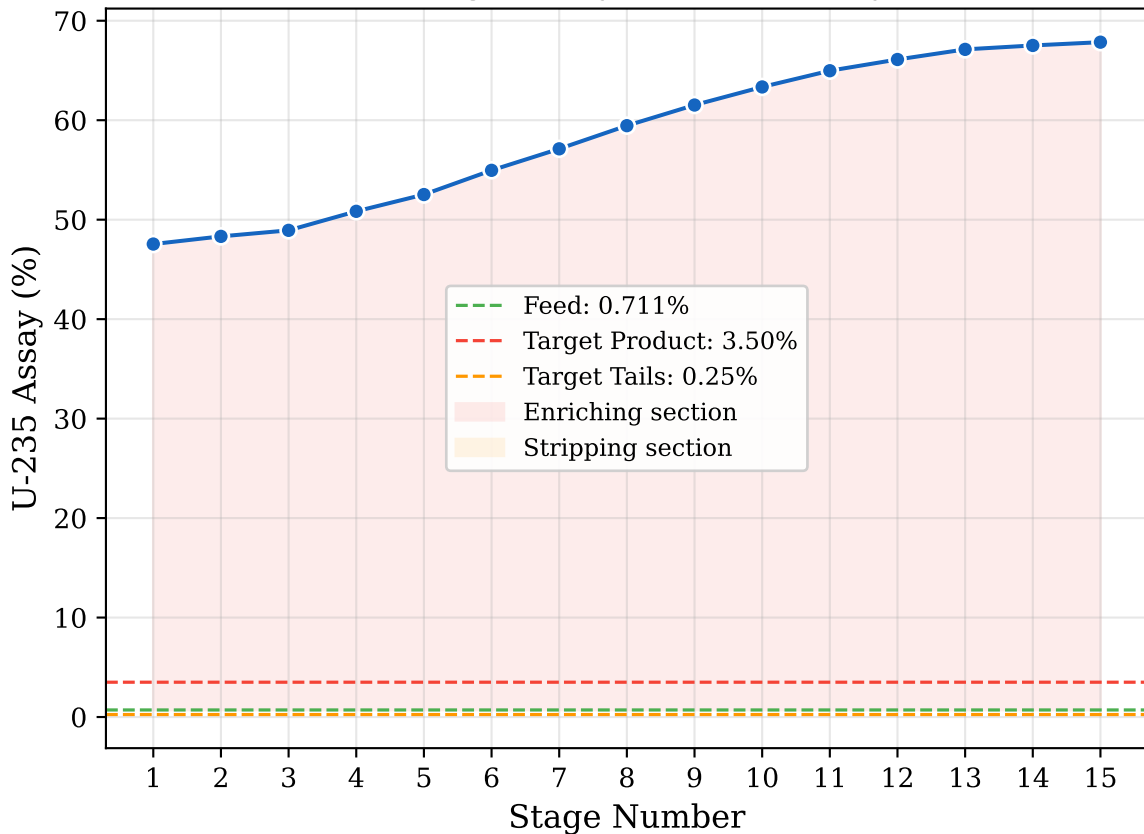
Gas Centrifuge Cascade Simulation

Enrichment Analysis Report

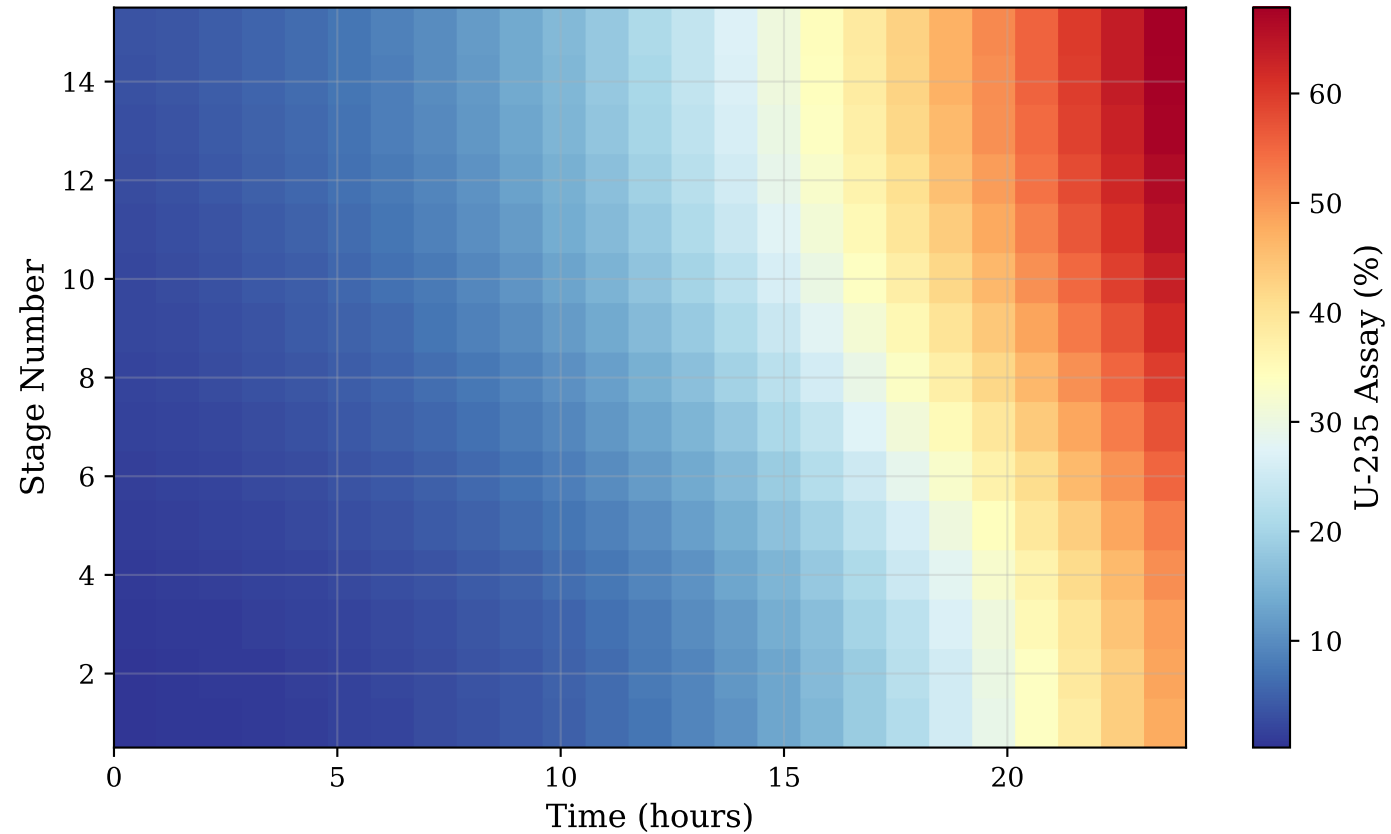
March 12, 2026

Generated by Nuclear Fuel Cycle Simulation Framework

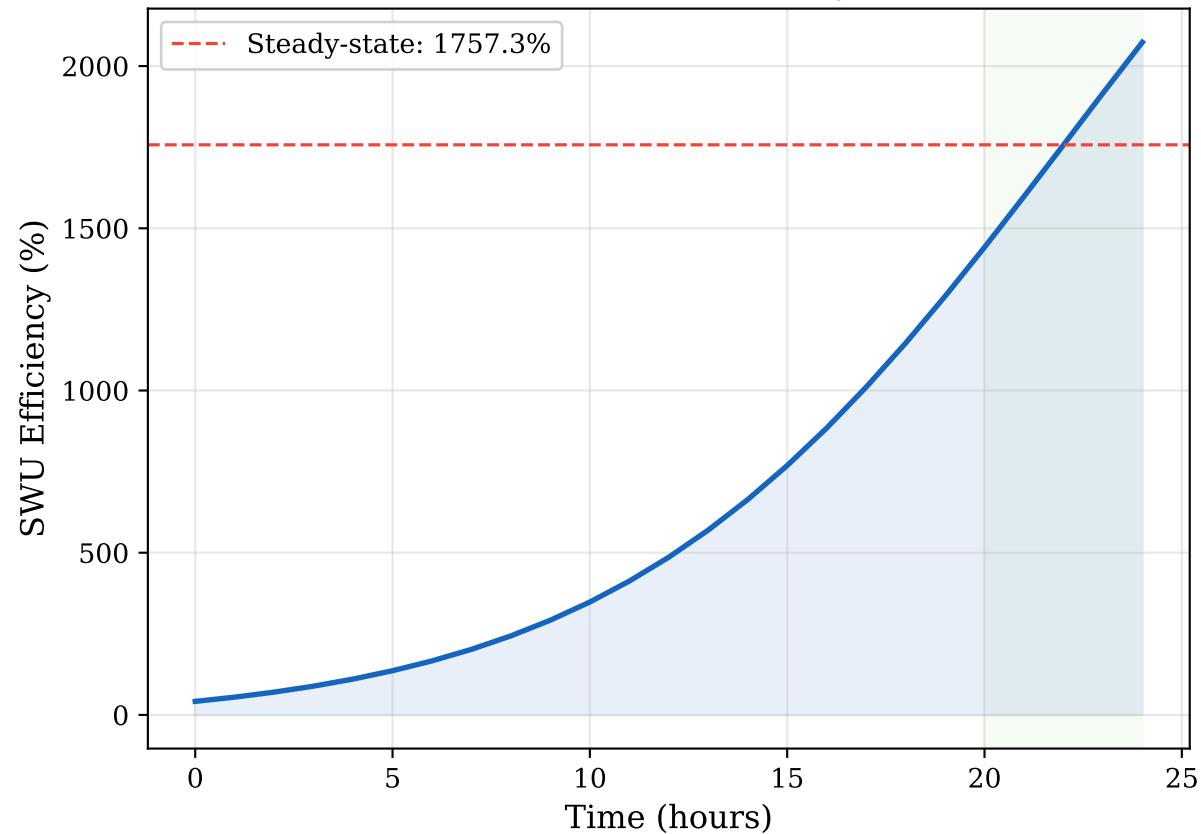
Cascade Stage Assay Profile (Steady State)



Dynamic Cascade Assay Evolution

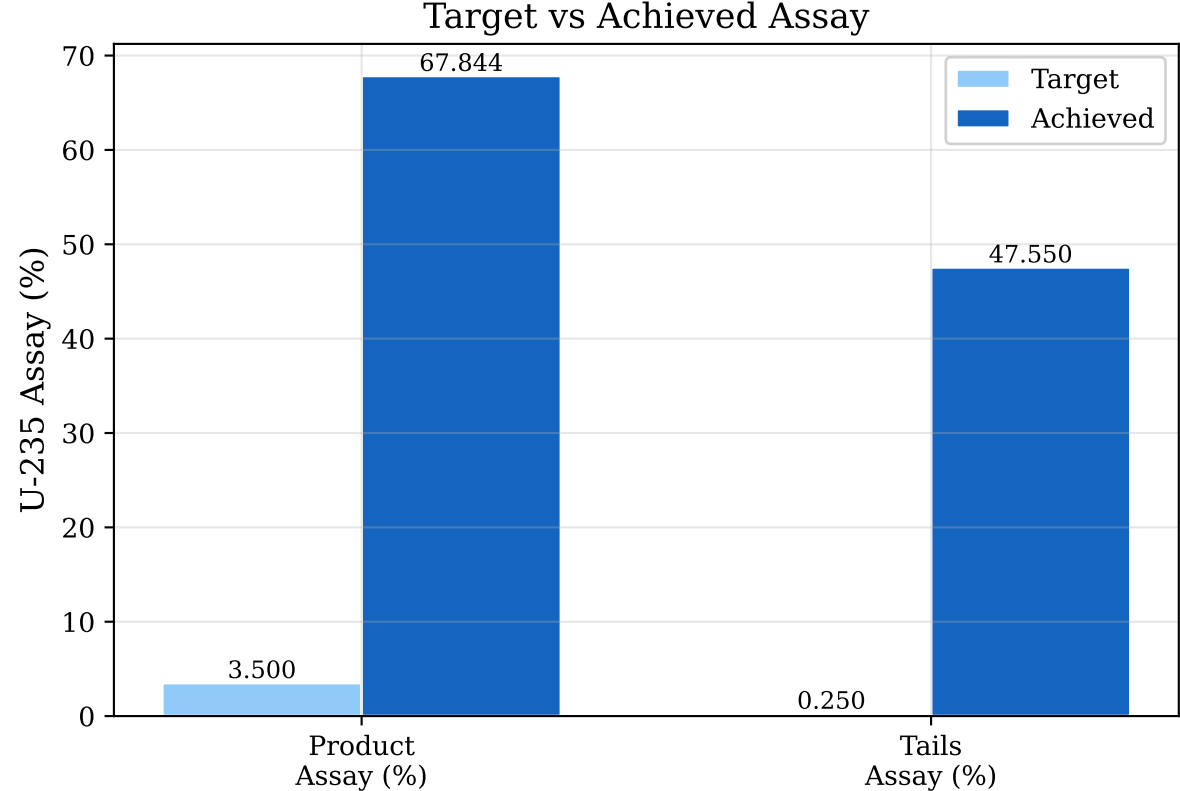


Separative Work Unit Efficiency Over Time



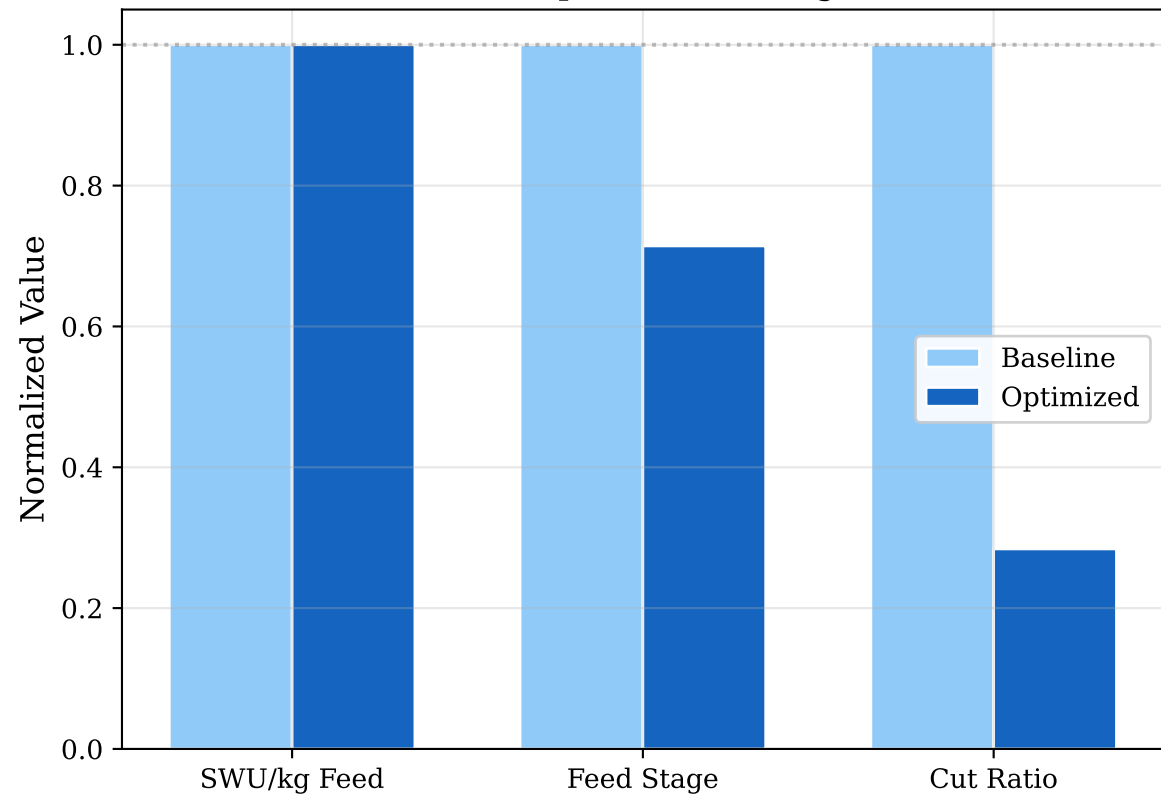
Gas Centrifuge Cascade Performance

Cascade Performance Metrics

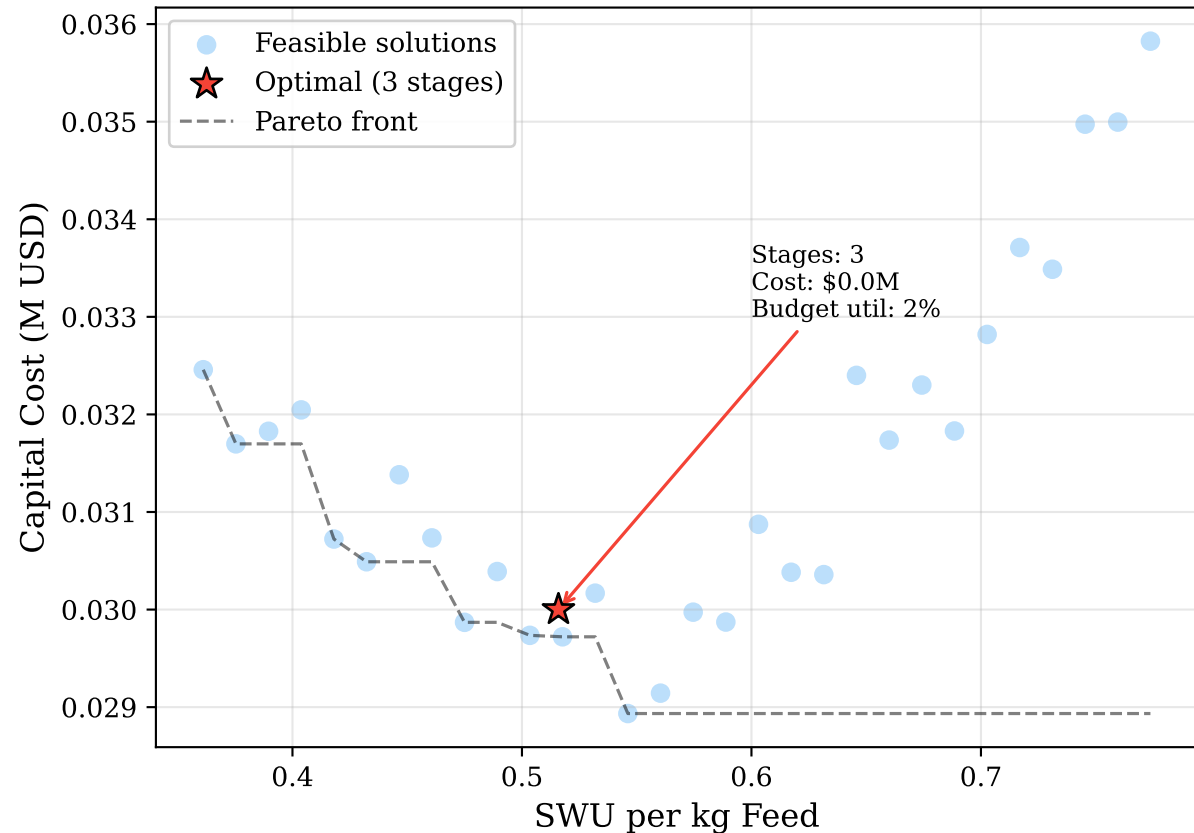


Metric	Value
Product Assay (achieved)	67.8444%
Tails Assay (achieved)	47.5497%
Separation Efficiency	2407.08%
Total SWU	1637.78
SWU per kg Feed	0.6824
Number of Stages	15
Machines	1,500

Baseline vs Optimized Configuration



Multi-Objective Optimization: SWU vs Cost



PCE vs Monte Carlo: Product Assay Prediction

367.6796 +/- 1190201662774253437064255221693037412111899250705563648.0000

Product Assay (%)

1.0

0.5

0.0

-0.5

-1.0

65.2314 +/- 19.5666

-- Target: 3.50%

PCE

Monte Carlo

1e54

Product Assay Uncertainty Distribution (PCE)

PCE distribution

95% CI: [-2332795259037536675395114188749429904331893193664626688.0000, 2332795259037536675395114188749429904331893193664626688.0000]%

Mean: 367.6796%

Target: 3.50%

